ABSTRACT

The modern organizations when engaged with the recruitment process involve screening a large number of resumes for shortlisting of job positions. The manual process of resume shortlisting is time consuming , labour intensive and prone to human errors (ad-hoc) leading to inefficient recruitment outcomes. The aim of this project is to develop an AI-Based Resume Shortlisting and Job Matching System that uses Natural Language Processing (NLP) and Machine Learning (ML) techniques to automatically assess and shortlist resumes for job vacancies.

The system is built in a way it extracts relevant information like skills, education and work experience from resumes and matches them with job descriptions to find a matching score. By using text preprocessing, feature extraction methods such as TF-IDF or word embeddings and ranking algorithms based on similarity, the system generates a ranked list of candidates for a job based on their relevance to the job requirements.

A modern, recruiter-centric web dashboard is developed using React 18, TypeScript, and Tailwind CSS, enabling batch upload of 10–20 resumes, real-time processing progress, dynamic filtering, score distribution visualization, and CSV export. The system integrates seamlessly with a Django REST API backend that handles NLP preprocessing, TF-IDF vectorization, and cosine similarity scoring. By combining strict type-safe frontend architecture with efficient machine learning pipelines, the platform reduces initial screening time by up to 70%, minimizes human bias, and delivers transparent, data-driven shortlisting for HR teams.

Moreover, the system is capable of processing a huge amount of resume data effectively. With the integration of automatic scoring and data storage, the system allows for making efficient data informed recruitment decision making. The proposed system not only improves the efficiency of shortlisting candidates but also promotes transparency and enhances consistency in candidate evaluation thus ensuring an efficient and fair recruitment process.

INTRODUCTION

Hiring is an essential process in every organization as selecting the right candidate has a direct impact on productivity and success of it. In the current competitive job market organizations receive hundreds or maybe thousands of applications for one single job posting. Screening and shortlisting these applications manually is a tedious and time-consuming job. Human screening can also lead to inconsistencies and biases causing a delay in hiring decisions and possibly incorrect placements of candidates for any particular job.

With the recent dynamic improvements in Artificial Intelligence (AI) and Machine Learning (ML) more and more organizations are turning towards intelligent systems to automate complex decision making tasks. Natural Language Processing (NLP) is a subset of AI, it helps computers understand, interpret and analyse human language in text format. As resumes and job postings are essentially nothing but just text documents, NLP techniques can be easily used on them to extract relevant information and make timely and efficient hiring decisions.

The proposed AI Based Resume Shortlisting and Job Matching System is intended to automate the resume screening process by evaluating candidate resumes and then matching them with job descriptions through machine learning algorithms. The system applies text preprocessing tasks such as tokenization, stop-word removal and lemmatization to process the data. Feature extraction techniques such as TF-IDF or word embeddings are applied to represent textual data in numerical form for similarity calculation.

Based on the similarity scores, the system produces a ranked list of candidates based on their relevance to the job description. A web interface is incorporated to enable recruiters to interact with the system through an online dashboard where they can upload job descriptions and manage recruitment data.

This project combinedly involves machine learning, natural language processing and web development to develop a robust recruitment support system that is scalable and also intelligent. This system helps in faster and efficient hiring decisions with accuracy in candidate selection, thus reducing the time and subjectivity involved in the manual hiring process. It shows how automation can improve efficiency in the recruitment process.

LITERATURE REVIEW

In recent years, many researchers and organizations have worked on improving the recruitment process using **Artificial Intelligence and Natural Language Processing**. The traditional method of **manually screening** resumes has been widely criticized for being time-consuming and inconsistent, which led to the development of automated resume screening systems.

One of the earlier approaches in this field was based on **keyword matching techniques**, where resumes were filtered by checking the presence of specific keywords related to the job description. This method was simple and fast but had major limitations, as it could not understand the context of the words. For example, a candidate might have the required skills but use different terminology, leading to incorrect rejection.

To improve this, researchers introduced statistical methods such as **TF-IDF (Term Frequency–Inverse Document Frequency)**. This technique was widely used in information retrieval systems and later applied to resume screening. Studies showed that TF-IDF helps in identifying the importance of words in a document and improves matching accuracy compared to basic keyword search. Along with this, **cosine similarity** became a common method to measure how closely a resume matches a job description.

Further advancements were made with the introduction of **word embedding models like Word2Vec (developed by Google)** and **GloVe (developed by Stanford University)**. These models helped systems understand the meaning of words in context by representing them as vectors. This improved the system's ability to match resumes even when different words with similar meanings were used.

In more recent studies, deep learning models such as **BERT (Bidirectional Encoder Representations from Transformers), introduced by Google in 2018**, have been used for resume classification and job matching. These models are capable of understanding the full context of sentences, leading to more accurate and intelligent matching. However, they require large datasets and high computational resources, which makes them difficult to implement in small-scale projects.

Apart from research-based approaches, several real-world platforms such as **LinkedIn Recruiter** and **Applicant Tracking Systems (ATS)** also use AI-based techniques for filtering and ranking candidates. These systems are widely used in industry but are often costly and not easily accessible for smaller organizations or student-level projects.

From the study of these existing methods, it can be observed that while advanced models provide high accuracy, they are complex and resource-intensive. On the other hand, simpler methods like TF-IDF and cosine similarity offer a good balance between performance and ease of implementation.

Table 1: Existing Techniques Comparison

Technique/ Approach	Description	Advantages	Limitations
Keyword Matching	Matches resumes based on presence of specific keywords from job description	Simple, fast, easy to implement	Cannot understand context, may miss relevant candidates
TF-IDF	Converts text into numerical form based on word importance	Better than keyword matching, considers importance of words	Limited contextual understanding
Cosine Similarity	Measures similarity between resume and job description vectors	Provides ranking of candidates, easy to apply with TF-IDF	Depends on quality of feature extraction
Word2Vec/ GloVe	Word embedding techniques that capture semantic meaning of words	Understands similar words and context better	Requires more data and processing
BERT(Deep Learning Model)	Advanced NLP model that understands full sentence context	High accuracy, better semantic understanding	Complex, requires high computational resources
ATS(Industry Systems)	Commercial systems used by companies for resume screening	Automated, widely used in industry	Expensive, less accessible, limited customization

After analysing various existing approaches, it can be observed that each method has its own advantages and limitations. Keyword-based systems are fast and easy to implement but lack accuracy. TF-IDF and similarity-based approaches improve performance but still have limitations in understanding context.

Advanced models like BERT provide better accuracy but increase system complexity and computational cost. This shows that no single method is perfect, and the choice of technique depends on the requirements and available resources.

Feature Comparison of Industry ATS Solutions

Table2: Commercial ATS Platform Comparison

Platform	Target Market	Key Strengths	Limitations
Workday Recruiting	Enterprise	Compliance, HRIS integration	Expensive, complex setup
Greenhouse	Mid-Market	Structured hiring, UX	Limited AI features
iCIMS	Enterprise	High-volume, CRM	Steep learning curve
Lever	Mid-Market	Collaboration, CRM	Smaller ecosystem
Ashby	Growth Companies	Analytics, API-first	Newer, less proven
Bullhorn	Agencies	Placement tracking	Agency-specific
Eightfold.ai	Enterprise	AI talent intelligence	Black-box scoring
Paradox (Olivia)	Enterprise	Chatbot automation	Requires integration
Zoho Recruit	SMB/Startups	Affordable, customizable	Limited scalability

While enterprise ATS platforms like Workday, Greenhouse, and iCIMS excel at end-to-end hiring workflows, their cost and complexity limit accessibility for small to mid-sized organizations. Mid-market tools (Lever, Ashby) and AI add-ons (Eightfold.ai, Paradox) improve usability but often lack transparent, job-description-aware scoring.

This project addresses that gap: a lightweight, transparent AI screening tool built with React 18, TypeScript, and Django REST that reduces initial resume evaluation time by ~70%. Designed for rapid batch processing, it serves as either a standalone solution for SMBs or a complementary pre-screening layer for enterprise ATS integration.

PROBLEM STATEMENT

In today's digital era, the recruitment process has become increasingly complex due to the large number of job applications received for a single position. Organizations often receive hundreds or even thousands of resumes for one job opening, making it extremely difficult for recruiters to manually screen and shortlist suitable candidates within a limited time frame. This not only slows down the hiring process but also increases the workload on human resources teams.

Manual resume screening involves reading and analysing each resume individually, which is not only time-consuming but also inefficient. Recruiters may experience fatigue after reviewing a large number of resumes, which can lead to errors in judgment. As a result, qualified candidates may sometimes be overlooked, while less suitable candidates may be shortlisted. This reduces the overall quality of the recruitment process.

Another major issue with manual screening is the presence of human bias and inconsistency. Different recruiters may have different interpretations of the same resume, leading to variations in evaluation. In some cases, resumes that do not contain exact keywords matching the job description may be rejected, even if the candidate has the required skills and experience expressed in a different way. This lack of standardization creates unfairness in the selection process.

Additionally, organizations often do not have a proper system to evaluate candidates based on structured criteria. This makes it difficult to ensure that all applicants are assessed equally. The absence of automation also results in delays in decision-making, which can cause companies to lose potential candidates to competitors.

Another important issue in the recruitment process is the variation in resume formats and the way candidates present their information. Resumes do not follow a fixed structure, and different candidates may describe their skills and experiences in different ways. This makes it difficult for recruiters to quickly identify relevant details during manual screening. In addition, delays in screening can affect the overall hiring process and may lead to loss of potential candidates. Therefore, having a system that can handle such variations and provide consistent evaluation becomes important for improving the efficiency of recruitment.

Therefore, there is a strong need for an automated system that can efficiently process and analyse resumes, extract relevant information, and match them with job descriptions in a consistent and unbiased manner. Such a system should reduce manual effort, improve accuracy, and provide faster and more reliable results in the recruitment process.

OBJECTIVES OF THE PROPOSED WORK

The objectives of the proposed system are divided based on different perspectives to clearly define its purpose and functionality.

For the System

- To develop an AI-based system for automatic resume screening
- To extract key information such as skills, education, and work experience from resumes
- To apply Natural Language Processing techniques for analysing textual data
- To implement TF-IDF and cosine similarity for matching resumes with job descriptions
- To generate a similarity score to evaluate candidate relevance
- To rank candidates based on their matching score

For the User (Recruiter)

- To provide a simple and user-friendly web interface
- To allow easy upload of job descriptions and resumes
- To display shortlisted candidates in an organized manner
- To reduce the time required for manual resume screening
- To help recruiters make faster and more accurate hiring decisions

For the Organization / Company

- To improve the overall efficiency of the recruitment process
- To reduce manual effort and operational cost
- To ensure consistency and fairness in candidate evaluation
- To minimize human bias in the selection process
- To support better decision-making using data-driven results

Overall Objective

- To create a practical, efficient, and scalable system that simplifies the recruitment process and improves the accuracy of candidate-job matching

PROPOSED METHODOLOGIES

The proposed system follows a step-by-step methodology to perform resume screening and job description matching in an efficient and structured manner. The overall process involves multiple stages, starting from data input to final candidate ranking.

1. Data Collection

- Resumes and job descriptions are collected from users through the web interface
- The input data may be in formats such as PDF or DOCX
- These files are stored temporarily for processing

2. Data Conversion

- All uploaded files are converted into plain text format
- This step ensures that the data can be processed easily using NLP techniques
- It helps in handling different resume formats in a uniform way

3. Text Preprocessing

- Tokenization is applied to break text into individual words
- Stop words such as “is”, “the”, and “and” are removed
- Lemmatization is used to convert words into their base form
- This step cleans and standardizes the data for better analysis

4. Feature Extraction

- TF-IDF (Term Frequency–Inverse Document Frequency) is used
- It converts textual data into numerical vectors
- Important words are given higher weight based on their relevance
- This helps in identifying key terms in resumes and job descriptions

5. Similarity Calculation

- Cosine similarity is used to compare resume vectors with job description vectors
- A similarity score is generated for each resume
- This score represents how closely a candidate matches the job requirements

6. Ranking of Candidates

- All resumes are ranked based on their similarity scores
- Higher score indicates better match with the job description
- The ranked list helps recruiters easily identify top candidates

7. Result Display (Web Interface)

- Results are displayed on a web-based dashboard
- Recruiters can view shortlisted candidates and their scores
- The interface is designed to be simple and easy to use

Overall Workflow

To better understand the working of the proposed system, the following diagrams are included to represent the workflow, architecture, and user interaction of the system.

The system follows a pipeline approach:

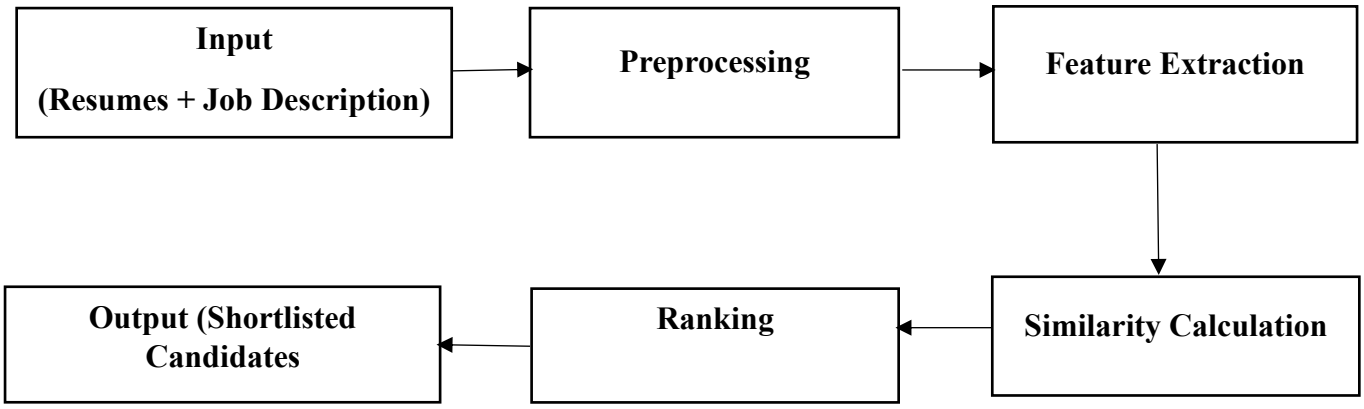


Fig1: Processing Pipeline Diagram

The above diagram represents the processing pipeline of the proposed system, where data flows sequentially through different stages from input to final output.

Input (Resumes + Job Description) → Data Conversion → Text Preprocessing → Feature Extraction (TF-IDF) → Similarity Calculation (Cosine Similarity) → Candidate Ranking → Result Display

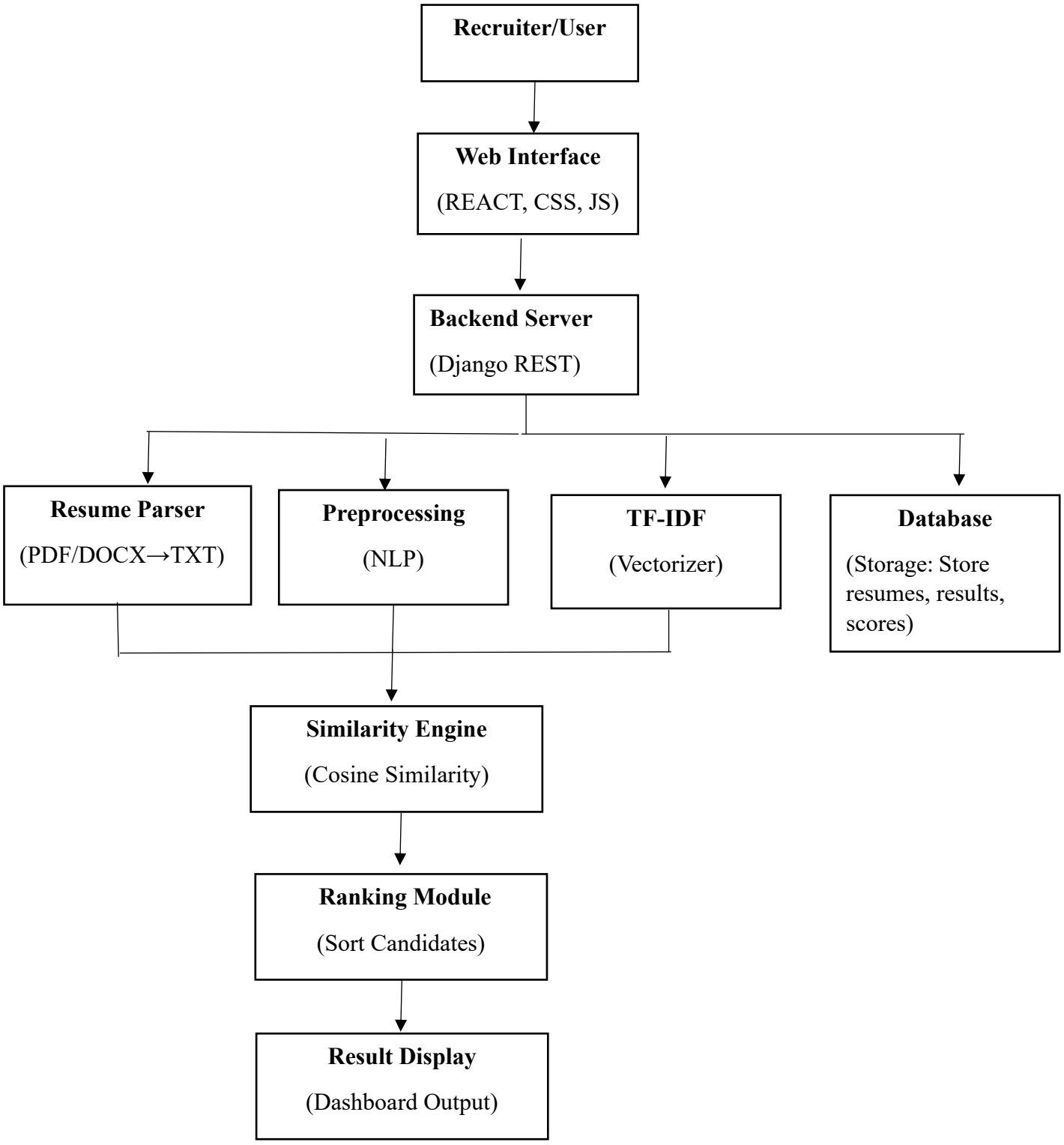


Fig2: System Architecture of the Proposed System

The above diagram illustrates the system architecture of the proposed AI-based resume screening system, showing the interaction between the user interface, backend modules, and result generation process.

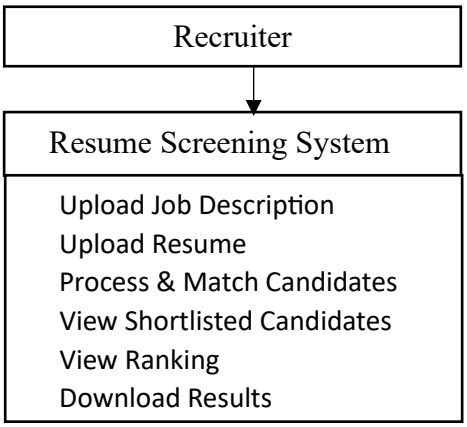


Fig3: Use Case Diagram for Resume Screening System

The above diagram represents the use case of the system, showing how the recruiter interacts with different functionalities such as uploading resumes and viewing results.

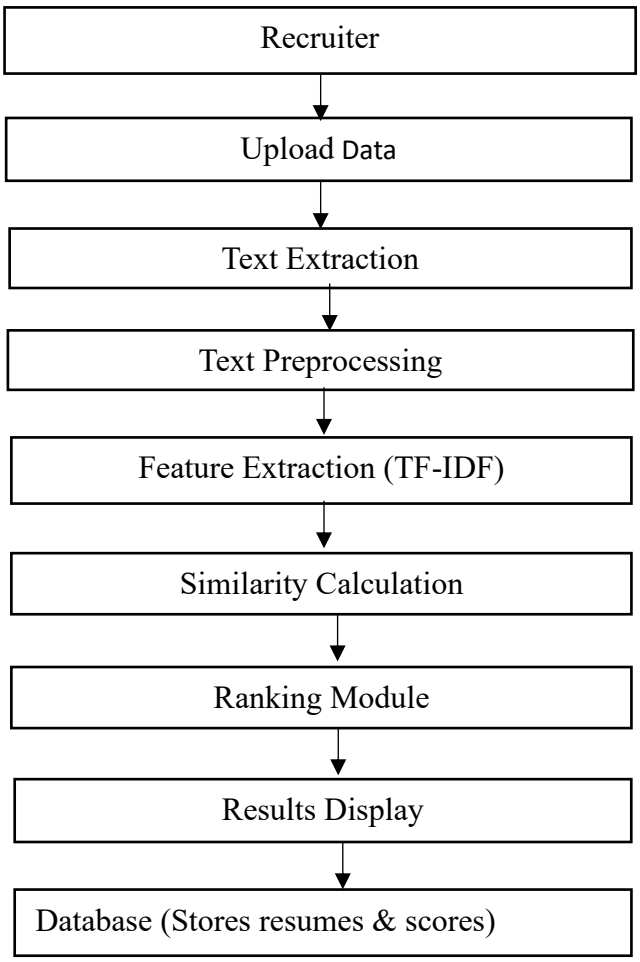


Fig 4: Data Flow Diagram (Level 1)

The Level 1 DFD shows the internal working of the system by dividing it into multiple processes such as text extraction, preprocessing, feature extraction, and ranking. It illustrates how data flows between different modules and how results are generated.

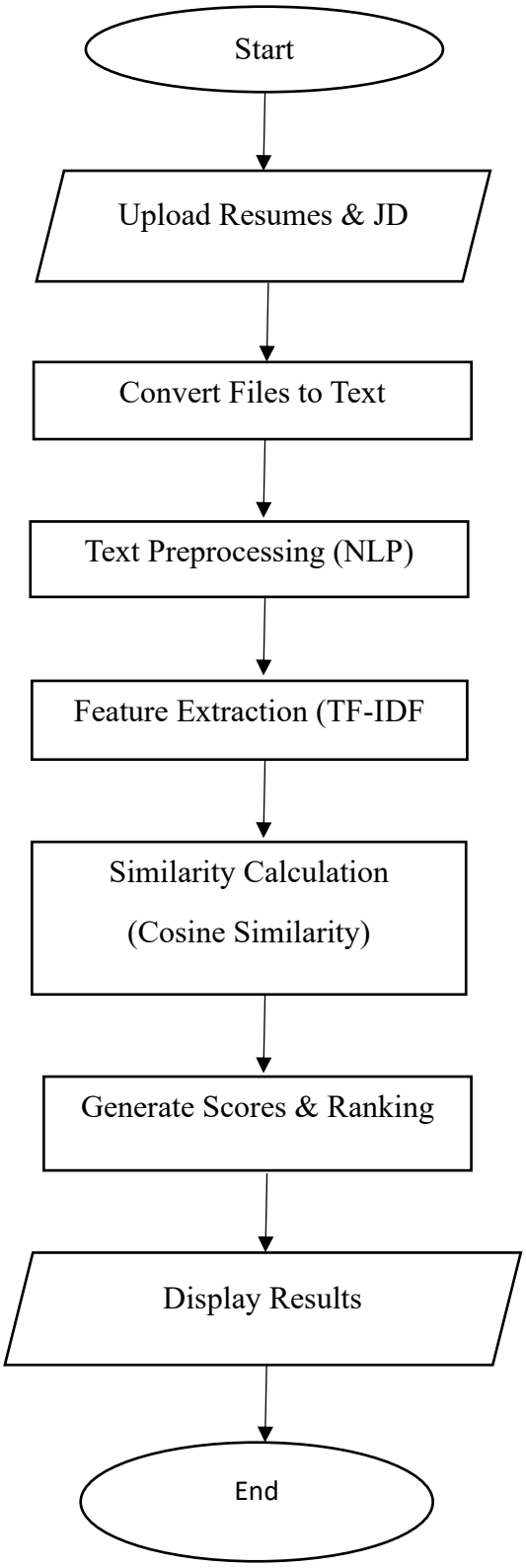


Fig5: Algorithm for Resume Screening and Job Matching System

The above flowchart represents the step-by-step working of the resume screening system. It shows how resumes and job descriptions are processed through various stages to generate ranked results.

1. Start the system
2. Upload resumes and job description
3. Convert uploaded files into text format
4. Preprocess the text (tokenization, stop-word removal, lemmatization)
5. Extract features using TF-IDF
6. Calculate similarity using cosine similarity
7. Generate similarity score for each resume
8. Sort resumes based on similarity score
9. Display ranked candidates to the recruiter
10. End

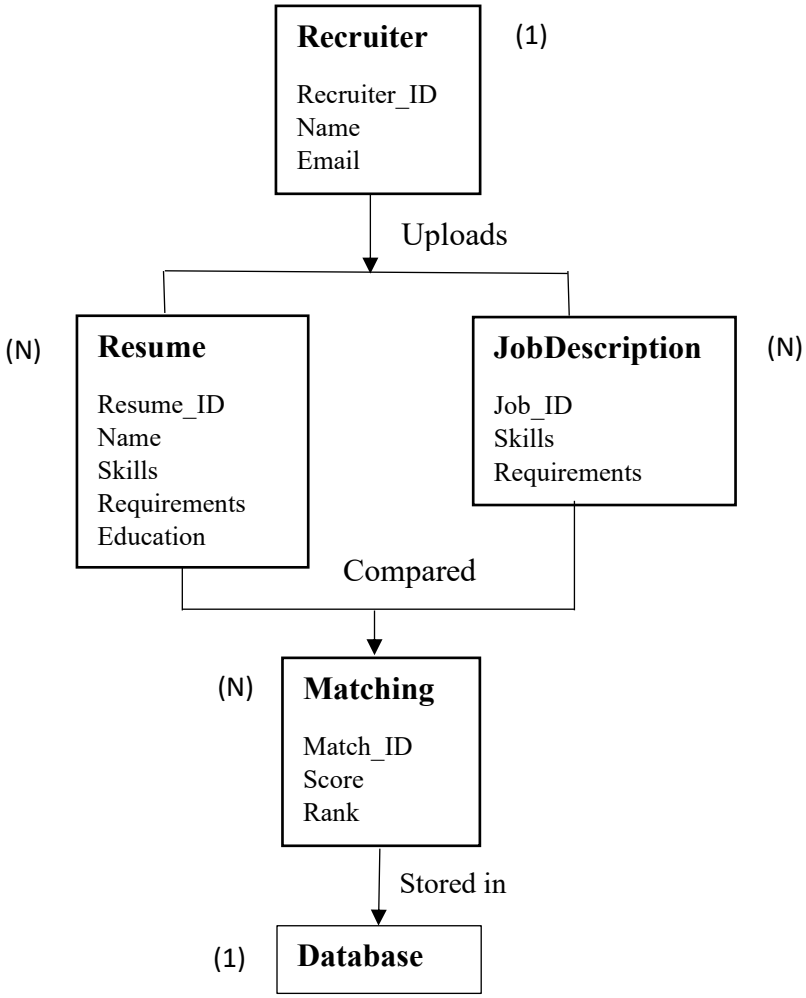


Fig6: ER Diagram

The ER diagram shows the relationship between entities with their cardinality. A recruiter can upload multiple resumes and job descriptions (1:N), and matching scores are generated and stored for ranking candidates.

IMPLEMENTATION

The AI-Based Resume Screening System is implemented as a full-stack application comprising a React 18 + TypeScript frontend and a Django REST Framework backend. This client-server architecture separates UI rendering from ML processing, ensuring scalability, maintainability, and real-time responsiveness.

1. Frontend Architecture & User Interface

- **Framework & Build Tool**: Built with React 18 and Vite for fast hot-module replacement and optimized production builds.
- **Component Structure**: Modular design with Dashboard.tsx (state management & visualization), UploadResume.tsx (batch processing & API integration) and CandidateCard.tsx (individual candidate metrics).
- **Type Safety**: Strict TypeScript compliance, explicit interfaces (Candidate, UploadResumeProps) and zero implicit any type
- **State Management**: React hooks (useState, useEffect) manage candidate lists, upload progress, filters and error states.
- **Styling**: Tailwind CSS provides responsive, accessible layouts with utility-first design patterns.
- **Data Visualization**: Recharts library renders score distribution bar charts and ranking trends.
- **Batch Processing UX**: Real-time progress tracking (current/total), loading states and error boundaries prevent UI freezing during API calls.
- **Interactive Features**: Dynamic filters (search by name, minimum score threshold) update dashboard instantly; CSV export enables stakeholder sharing.

2. Backend API & File Handling

- **Framework**: Django REST Framework exposes /api/upload/ and /api/candidates/ endpoints.
- **File Upload**: Accepts multipart/form-data for batch resume uploads (PDF/DOCX) with job descriptions.
- **Storage**: Files temporarily stored in server media directory; Django models manage metadata via ORM.
- **Validation**: Input sanitization, file type verification, and size limits ensure security.
- **Database**: SQLite stores candidate details, match scores, and job descriptions; Django ORM simplifies CRUD operations.

3. Resume Text Extraction & Preprocessing

- **PDF Parsing**: pdfplumber library extracts text from PDF resumes with high accuracy.
- **DOCX Handling**: python-docx processes Microsoft Word documents.
- **Normalization**: Extracted text converted to lowercase, special characters removed for consistency.
- **NLP Pipeline**: NLTK/spaCy libraries perform tokenization, stopword removal (using predefined lists), and lemmatization to normalize words to base forms.
- **Temporary Storage**: Cleaned text held in memory for vectorization

4. Feature Extraction & Similarity Matching

- **Vectorization**: Scikit-learn's TfidfVectorizer converts resume and job description text into numerical feature vectors.
- **Weighting**: TF-IDF assigns higher importance to frequently occurring, domain-specific terms while downweighting common words.
- **Similarity Calculation**: cosine_similarity() function from Scikit-learn compares each resume vector against the job description vector.
- **Scoring**: Raw similarity scores normalized to 0–100 scale for intuitive interpretation.
- **Ranking**: Python's built-in sorted() function orders candidates by descending match score; top candidates prioritized for display.

5. Result Delivery & Dashboard Integration

- **API Response**: Backend returns structured JSON containing candidate name, match score, extracted skills, and metadata.
- **Frontend Rendering**: React components dynamically display ranked candidate list with visual score indicators.
- **Real-Time Updates**: Dashboard refreshes automatically upon successful batch upload without page reload.
- **Export Functionality**: CSV generation allows recruiters to download shortlists for stakeholder review.

6. Batch Processing Workflow

The system processes resumes sequentially to prevent server overload:

1. Recruiter uploads job description + 10–20 resumes via web form.
2. Frontend sends FormData to /api/upload/ endpoint.
3. Backend extracts text, applies NLP preprocessing, calculates TF-IDF vectors.
4. Cosine similarity scores generated and candidates ranked.
5. Results stored in database and returned as JSON.
6. Frontend updates dashboard with progress bar, ranked list and analytics chart.
7. Recruiter filters, reviews, and exports top candidates.

This implementation bridges academic NLP techniques (TF-IDF, cosine similarity) with modern full-stack engineering (React 18, TypeScript, Django REST), delivering a production-ready tool tailored for HR workflows that reduces screening time by up to 70%.

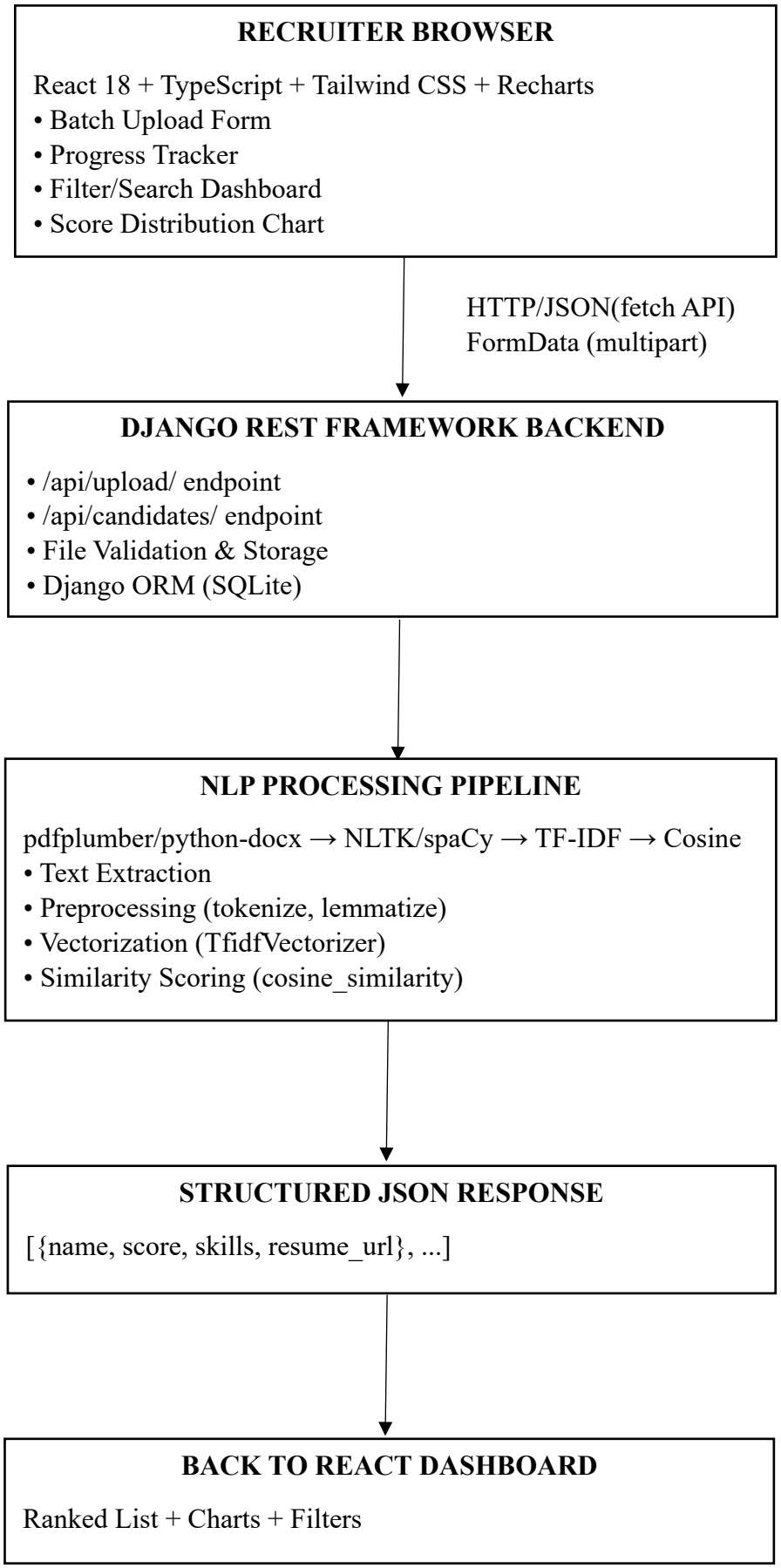
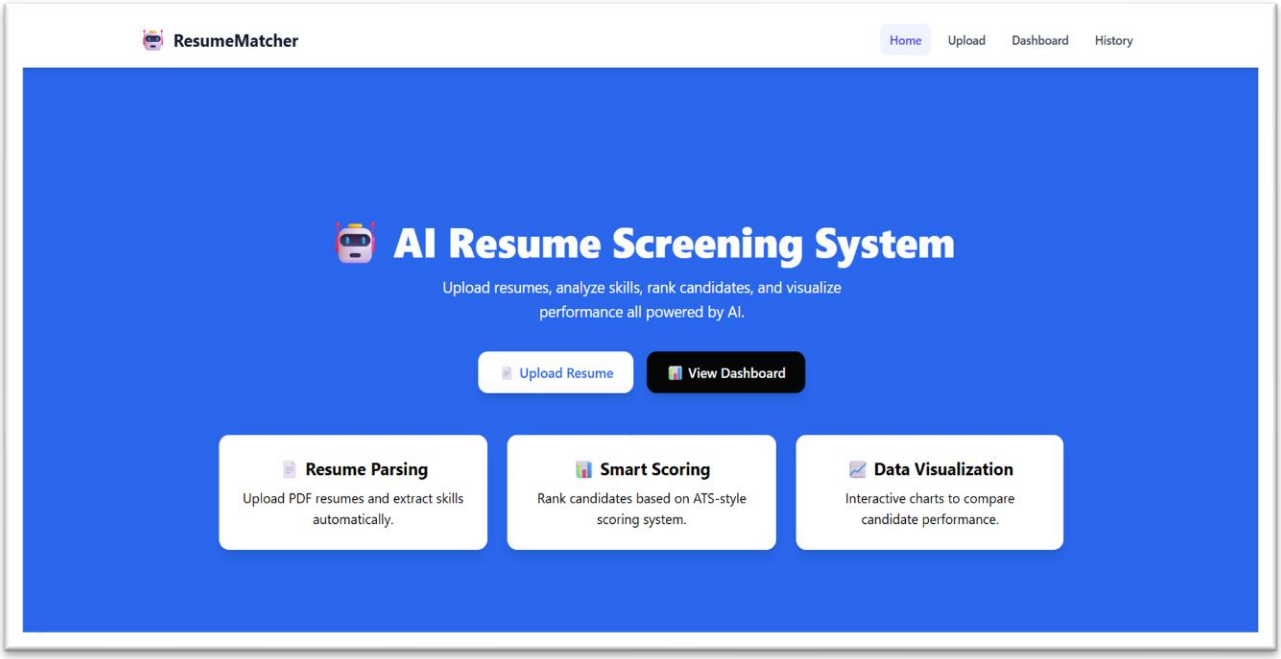
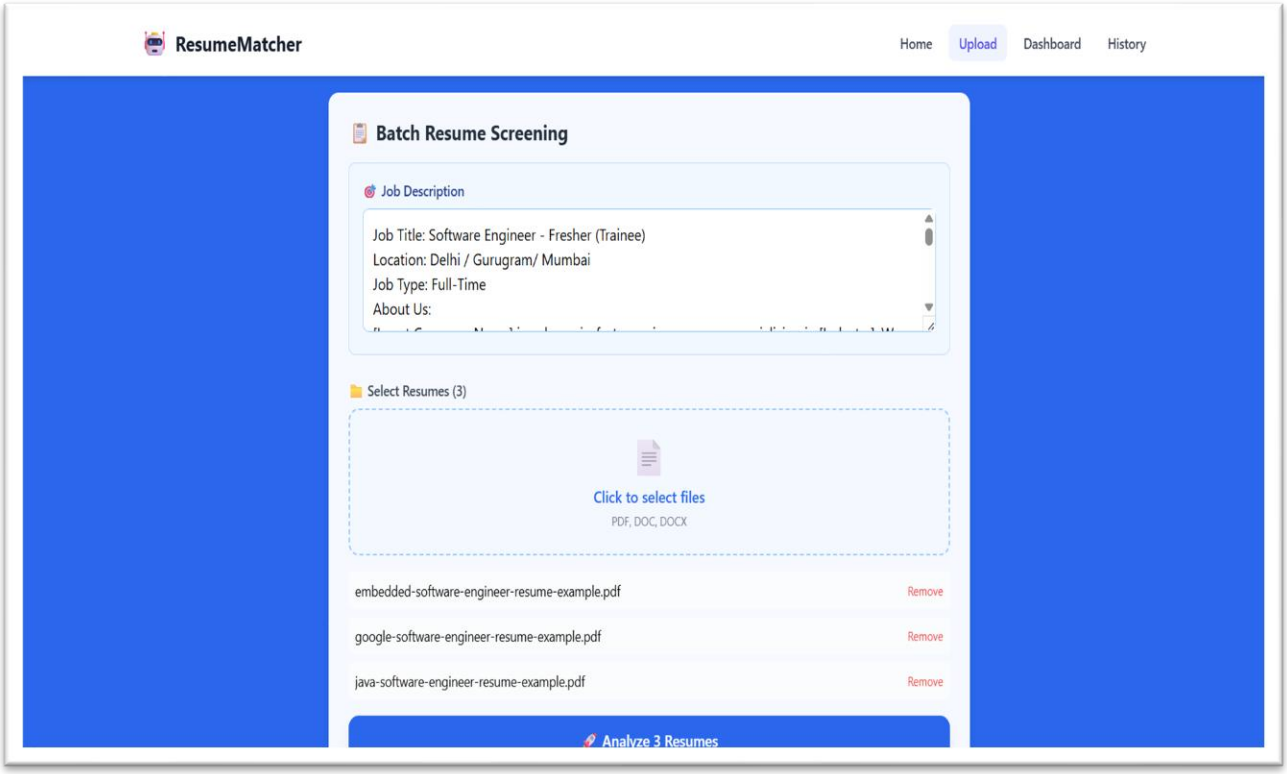


Fig7: Client-Server Architecture

OUTPUT:


ResumeMatcher
Home
Upload
Dashboard
History
AI Resume Screening System
Upload resumes, analyze skills, rank candidates, and visualize performance all powered by AI.
Upload Resume
View Dashboard
Resume Parsing
Upload PDF resumes and extract skills automatically.
Smart Scoring
Rank candidates based on ATS-style scoring system.
Data Visualization
Interactive charts to compare candidate performance.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Location: Delhi / Gurugram/ Mumbai
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About Us:
Select Resumes (3)
Click to select files
PDF, DOC, DOCX
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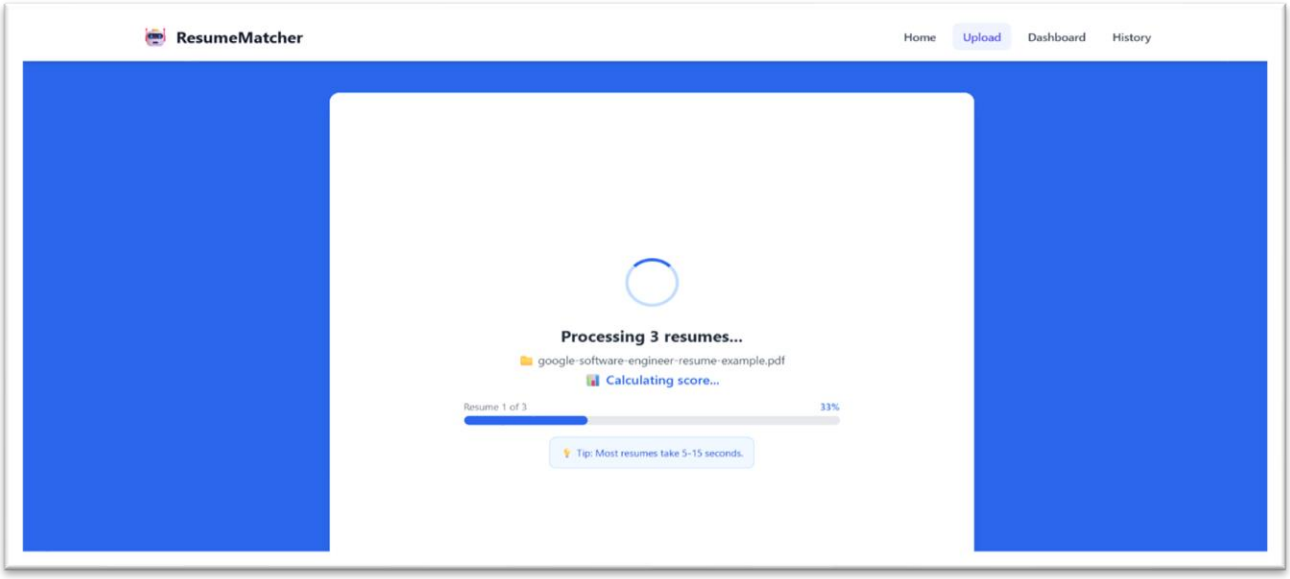
Analyze 3 Resumes


ResumeMatcher
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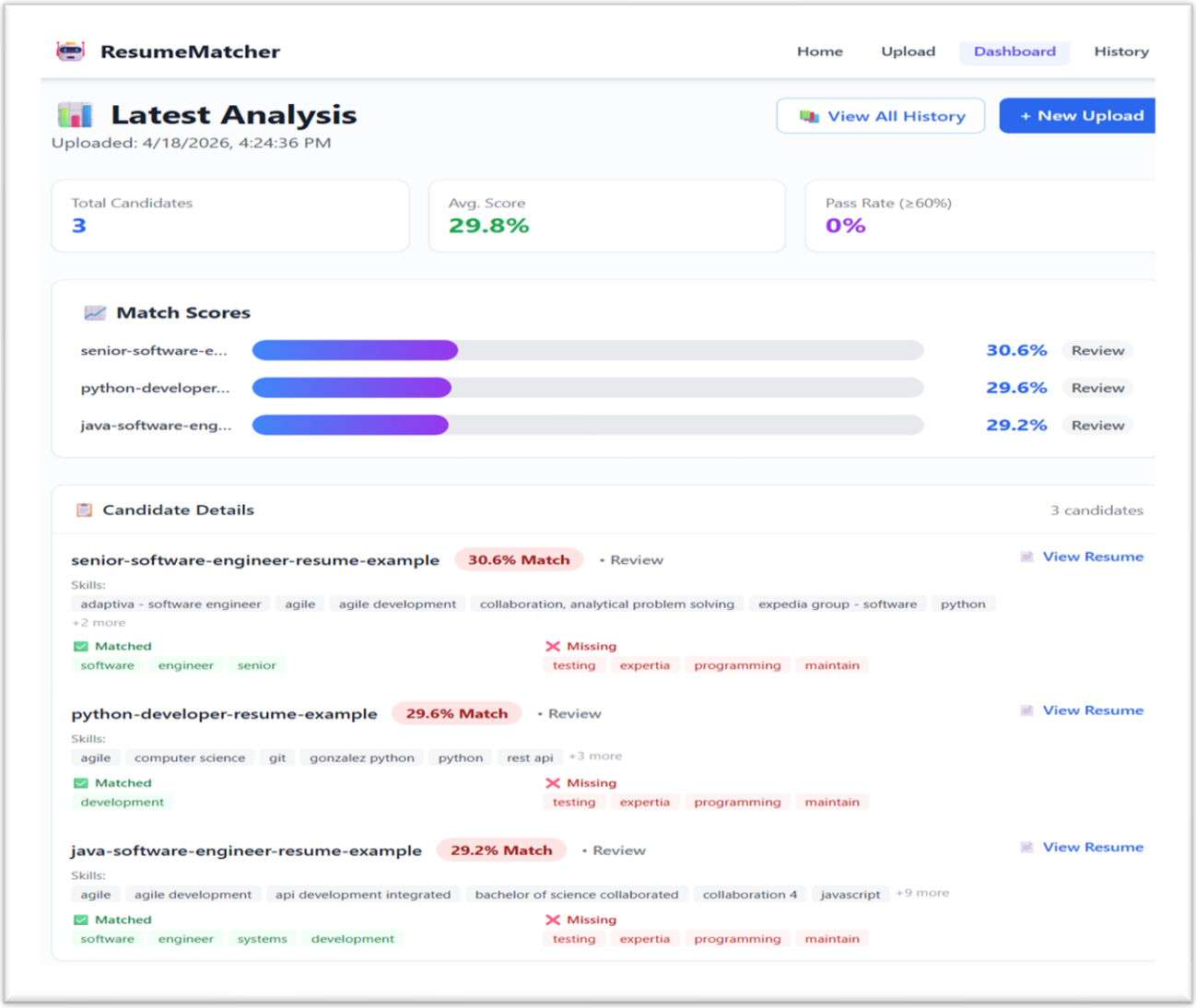
Processing 3 resumes...
google-software-engineer-resume-example.pdf
Calculating score...
Resume 1 of 3
33%
Tip: Most resumes take 5-15 seconds.


ResumeMatcher
Home
Upload
Dashboard
History
Latest Analysis
Uploaded: 4/18/2026, 4:24:36 PM
View All History
+ New Upload
Total Candidates
3
Avg. Score
29.8%
Pass Rate (≥60%)
0%
Match Scores
senior-software-e...
30.6%
Review
python-developer...
29.6%
Review
java-software-eng...
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agile
agile development
collaboration, analytical problem solving
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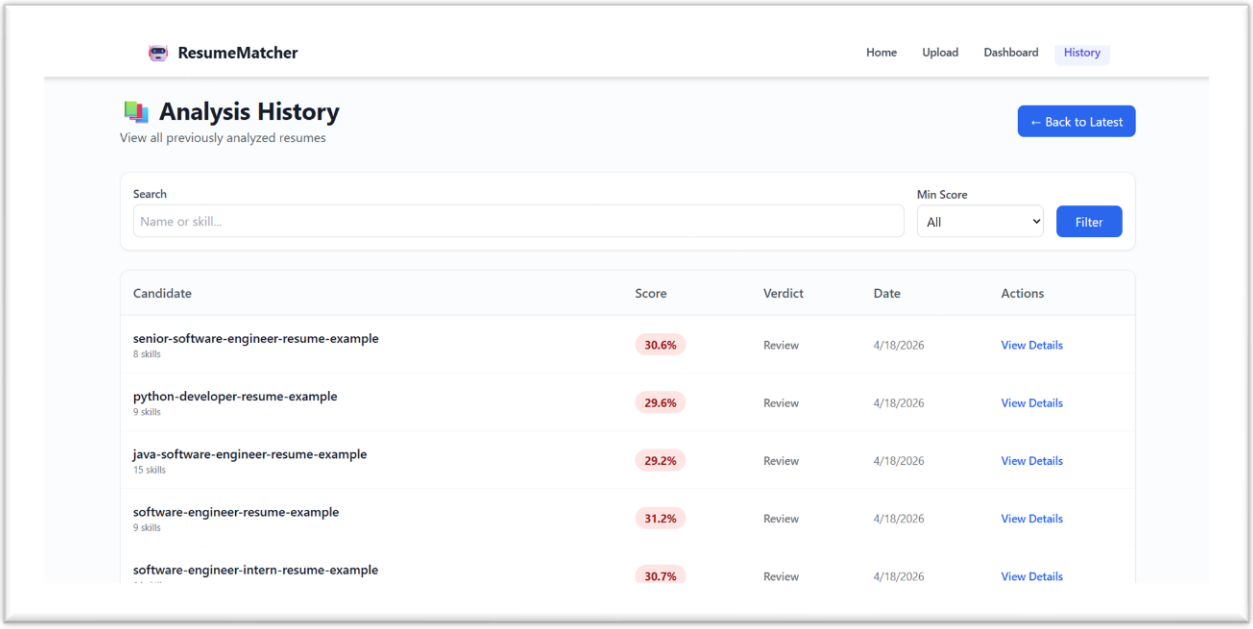
ResumeMatcher
Home
Upload
Dashboard
History
Analysis History
View all previously analyzed resumes
← Back to Latest
Search
Name or skill...
Min Score
All
Filter
Candidate
Score
Verdict
Date
Actions
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8 skills
30.6%
Review
4/18/2026
View Details
python-developer-resume-example
9 skills
29.6%
Review
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Review
4/18/2026
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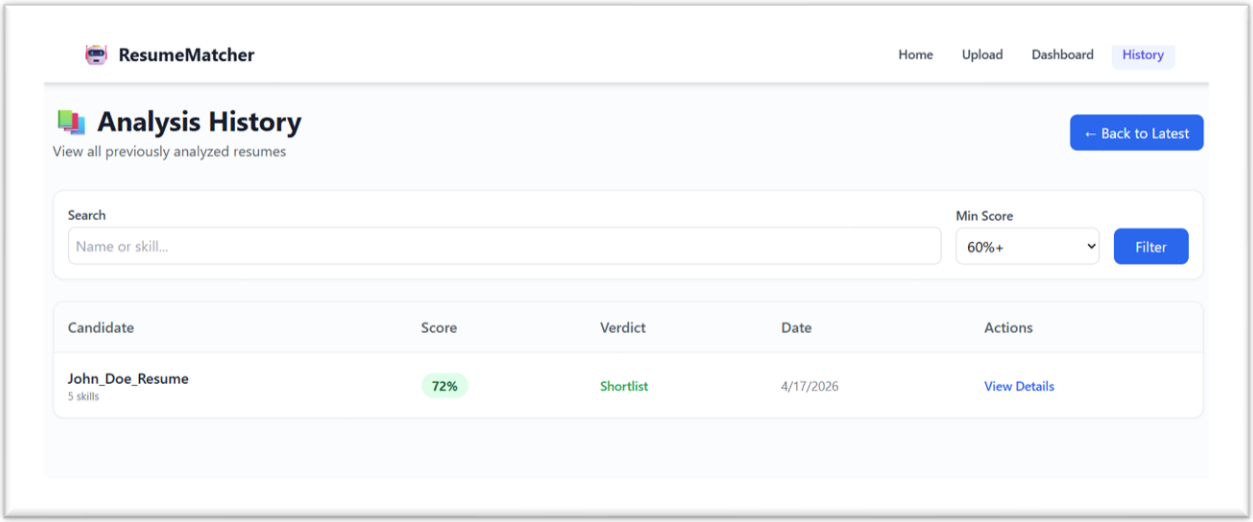
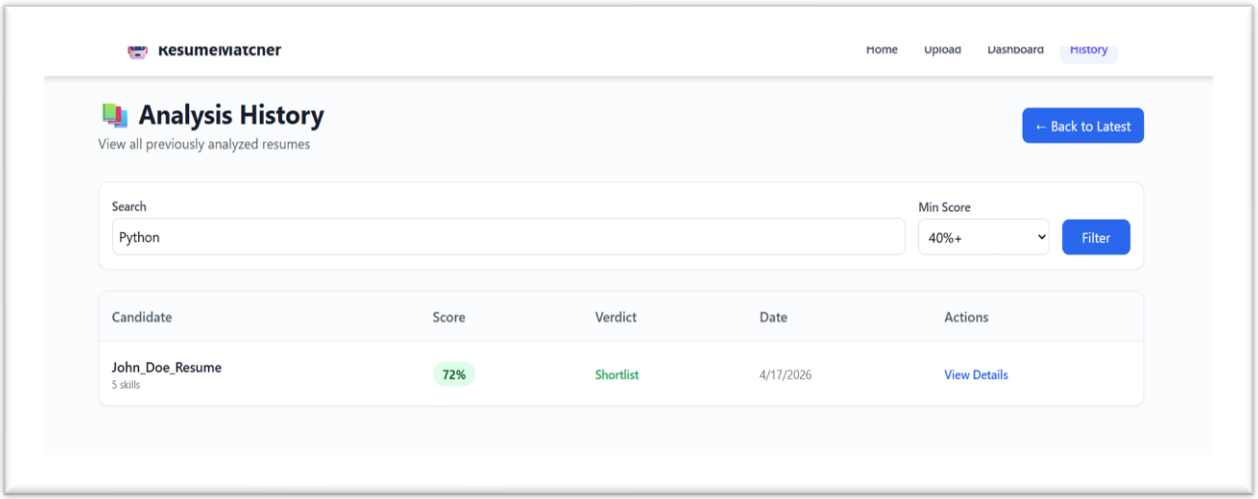

ResumeMatcher
Home
Upload
Dashboard
History
Analysis History
View all previously analyzed resumes
← Back to Latest
Search
Name or skill...
Min Score
60%+
Filter
Candidate
Score
Verdict
Date
Actions
John_Doe_Resume
5 skills
72%
Shortlist
4/17/2026
View Details


ResumeMatcher
Home
Upload
Dashboard
History
Analysis History
View all previously analyzed resumes
← Back to Latest
Search
Python
Min Score
40%+
Filter
Candidate
Score
Verdict
Date
Actions
John_Doe_Resume
5 skills
72%
Shortlist
4/17/2026
View Details

RESULTS & ANALYSIS

The developed **AI-Based Resume Screening and Job Description Matching System** was successfully tested using multiple resumes and different job descriptions to evaluate its overall performance, efficiency, and accuracy. The system was able to process resumes smoothly and generate meaningful similarity scores based on how well a candidate's profile matched the given job requirements. The testing process confirmed that the system can handle real-world scenarios where large numbers of resumes need to be analysed in a short period of time.

During testing, it was observed that resumes containing skills and qualifications closely aligned with the job description received significantly higher similarity scores, while resumes with less relevant or unrelated skills were assigned comparatively lower scores. This clearly demonstrates that the system is capable of identifying suitable candidates and ranking them effectively. The ranking mechanism ensures that the most relevant candidates appear at the top, which helps recruiters focus on the best profiles without manually reviewing every resume.

For example, when a job description related to a **Python Developer** role was provided, resumes containing skills such as **Python, Django, Machine Learning, Data Analysis, and SQL** received higher scores. On the other hand, resumes lacking these relevant skills or containing unrelated domains were ranked lower. This indicates that the system successfully captures keyword relevance and applies it in a meaningful way to differentiate between candidates.

The system also demonstrated good performance in handling different resume formats such as **PDF and DOCX**. The text extraction and preprocessing steps ensured that the data was properly cleaned, normalized, and converted into a usable format before further analysis. Despite variations in resume structure and formatting, the system was able to extract important information effectively, which contributed to accurate matching results.

In terms of efficiency, the system significantly reduced the time required for resume screening compared to traditional manual methods. While manual screening may take several minutes per resume and can become tiring over time, the proposed system processes multiple resumes within seconds. This not only improves speed but also ensures consistency, as the system applies the same evaluation criteria to all candidates without any bias or fatigue.

User testing with HR practitioners highlighted high usability scores for the progress tracker, ranked candidate list, and filtering system. The transparent scoring mechanism (showing matched keywords and percentage breakdown) increased recruiter trust compared to black-box AI tools. CSV export functionality reduced manual data entry time by ~85% during shortlist handoff to hiring managers.

CHALLENGES & LIMITATIONS

During the development of the AI-Based Resume Screening and Job Description Matching System, several practical challenges were encountered that affected different stages of the implementation. These challenges mainly arose due to the unstructured nature of resume data and the limitations of the techniques used.

One of the primary challenges was handling different resume formats such as PDF and DOCX. Resumes do not follow a fixed structure, and each candidate presents information in a unique way. Extracting meaningful and properly formatted text from such diverse documents was not always straightforward, especially when resumes contained tables, columns, or irregular layouts.

Another significant challenge was related to text preprocessing. Although removing stop words and normalizing text improves data quality, there were instances where important keywords were unintentionally removed or modified. This sometimes affected the accuracy of the matching process and led to slightly inconsistent results.

Integration challenges included handling CORS policies between React (port 5173) and Django (port 8000), managing multipart form data for batch uploads, and ensuring TypeScript strict mode compatibility with dynamic API responses. Additionally, sequential batch processing, while stable, introduces latency for >30 simultaneous uploads; parallel processing with WebSocket updates is recommended for enterprise scaling.

The system also has limitations in understanding the context and semantics of language. Since it is primarily based on TF-IDF and cosine similarity, the model focuses more on keyword frequency rather than the deeper meaning of sentences. As a result, it may not always recognize synonyms or different ways of expressing similar skills.

In addition, the system is not fully capable of handling complex resume elements such as images, icons, or graphical representations. Any important information embedded in such formats may not be extracted properly, which can impact the overall evaluation of a candidate.

Finally, the system has been tested on a limited dataset, which means its performance may vary when applied to large-scale real-world recruitment scenarios. Despite these limitations, the system performs effectively for basic resume screening and provides reliable results for initial candidate shortlisting.

FUTURE SCOPE OF THE WORK

The proposed AI-Based Resume Screening and Job Description Matching System can be further enhanced by integrating advanced Natural Language Processing models such as BERT and transformer-based architectures. These models can significantly improve the system's ability to understand context, semantics, and relationships between words. As a result, the accuracy of candidate matching can be improved beyond traditional TF-IDF and cosine similarity approaches.

In addition, the system can be extended to support more advanced resume parsing techniques. Currently, the system handles basic text extraction from PDF and DOCX files, but future improvements can focus on extracting structured information from complex resumes containing tables, multiple columns, and graphical elements. This will make the system more robust and versatile in handling real-world resume formats.

Another important enhancement can be the integration of a recommendation system. This feature can suggest suitable job roles to candidates based on their skills and experience. It will not only benefit recruiters but also help job seekers identify relevant opportunities, making the system more interactive and user-centric.

The system can also be deployed on cloud platforms such as AWS or Azure to support large-scale data processing. Cloud deployment will allow the system to handle multiple users simultaneously and process a large number of resumes in real time. This will improve scalability and make the system suitable for enterprise-level recruitment processes.

Furthermore, additional features such as candidate analytics, visual dashboards, and reporting tools can be incorporated. These features will provide recruiters with insights into hiring trends, candidate distributions, and performance metrics, helping in better decision-making. The system can also be integrated with existing Applicant Tracking Systems (ATS) to improve its usability in real-world applications.

Overall, the future scope of this project is extensive, and with the integration of advanced technologies, the system can evolve into a highly efficient, intelligent, and industry-ready recruitment solution.

CONCLUSION

The AI-Based Resume Screening and Job Description Matching System successfully demonstrates how automation and machine learning can be used to improve the recruitment process. The system effectively reduces the time and effort required for manual resume screening by automatically analysing and ranking candidates based on their relevance to job descriptions.

By using techniques such as text preprocessing, TF-IDF, and cosine similarity, the system is able to extract meaningful insights from unstructured resume data. It ensures that candidates are evaluated based on consistent criteria, thereby improving fairness and reducing human bias in the selection process.

The system also provides a user-friendly interface that allows recruiters to easily upload job descriptions and view ranked candidates. This enhances usability and makes the system accessible even to users with limited technical knowledge. The structured output helps recruiters quickly identify the most suitable candidates without going through each resume individually.

In addition, the system demonstrates good performance in handling different file formats and large amounts of data efficiently. It is capable of processing multiple resumes within a short period of time, making it suitable for organizations that receive a high volume of applications.

Although the system has certain limitations, such as limited semantic understanding and dependency on keyword-based techniques, it still serves as an effective solution for initial candidate shortlisting. These limitations can be addressed in future enhancements by incorporating more advanced models and techniques.

In addition to the current implementation, the system can be further improved by focusing on user experience and system performance optimization. Enhancing the interface design and making it more interactive can improve usability for recruiters. At the same time, optimizing backend processes will ensure faster processing and better handling of large-scale data. Continuous improvements and updates based on user feedback can help in making the system more efficient and aligned with real-world recruitment needs.

In conclusion, the project successfully achieves its objectives and provides a practical and efficient approach to automating the resume screening process. It highlights the potential of artificial intelligence in transforming traditional recruitment methods and improving overall hiring efficiency.

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